The Pleistocene-Holocene Transition in Tagua-Tagua, central Chile

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Abstract

Paleohydrological reconstructions from the Chilean Altiplano document abrupt moisture fluctuations during the last millennia. Although the end of the mid Holocene aridity and the onset of more humid conditions between 6–4 ka has been identified in numerous regional marine and terrestrial sites, the timing of late Holocene dry and humid phases shows large regional variability. Laguna Miscanti and Laguna Miñiques (23° 43'S – 67° 46'W, 4140 m asl) are topographically closed, but connected by surface outflow from Miscanti. Sedimentological and geochemical indicators from two new cores show large facies changes, i.e. higher carbonate and evaporite deposition during more arid periods and increased organic productivity (both algal and macrophyte) during more humid phases. As in most Andean lakes located in volcanic settings, large 14C reservoir effects occur complicating 14C dating, so the age models include 210Pb and U/Th dating. In spite of dating uncertainties, both lakes show similar patterns during the last millennium. A humid phase in Laguna Miscanti prior to ca 1200 CE is coherent with rodent middens and geomorphological features indicative of a major pluvial/recharge event at lower altitudes (Atacama Desert) during the Medieval Climate Anomaly (ca 800 - 1300 CE). The LIA (1300 – 1850 CE) is characterized by several arid/humid cycles and the last century by a productivity increase. The hydrological changes observed during the last millennium illustrate the complex dynamics of recent climate evolution over the high altitude Andean plateau. Discrepancies between the northern and southern Altiplano records and with intermediate latitudes (Central Chile) records may reflect contrasting responses to external forcing (Westerlies versus South American Monsoon dynamics) along different climatic zones.

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1. Introduction

2. Study site and regional setting

3. Materials and methods

3.1. Sedimentological facies and elemental geochemistry

In November 2019, we recovered a sedimentary sequence (~2.8 m deep) in an area of 15 x 10 cm following the archaeological strata of the profile exposed on the western wall of a 1x2 m excavation in the ALTT (Tagua-Tagua 3, Figure 1). Sedimentary facies were defined and characterized based on macroscopic description (color, bedding features, grain size, lithology and sedimentary textures). Sediment samples —between 5 to 10 g of material— were obtained at 1-cm intervals. Samples were sent to the laboratories of Pyrenees Ecology Institute (IPE-CSIC) and the Andalusian Institute Earth Sciences Institute (IACT-CSIC) in Spain for geochemical, pollen and isotopic analysis. Other samples group were sent to the Archaeobotany and Palaeoecology Laboratory of the University of Exeter for the processing and extraction of diatom and phytolith.

Subsamples were taken every 2 cm for analysed of total carbon (TC), total inorganic carbon (TIC), total organic carbon ($TOC = T - TIC$), total nitrogen (TN) and total sulfur (TS) in a LECO SCN furnace (IPE–CSIC). The grain-size frequency distributions (between 0.02 um and 2000 um) were measured in triplicate using a Malvern Mastersizer 2000 APA2000 grain-size analyzer with Hydro 2000G AWA2000. To determine the grain–size, the samples were pre-treated with 10% H2O2 and 10% HCl to remove organic matter and carbonates, respectively. Robust End-member mixing analysis (EMMA) was performed using the generic deterministic EMMA protocol slightly modified (R package EMMAgeo, version: 0.9.8. Dietze et al 2012; Dietze and Dietze 2019) to infer the sediment sources, transport and depositional processes.

3.2. Diatoms analysis

Approximately 1 g of dry sediment every 2 cm were processed followingthe Queen's University methods. Clastic sediments were removed using a sodium polytungstate treatment. Permanent slides were prepared with Naphrax (Battarbee et al., 2002). Diatoms were identified and quantified under a trinocula Carl Zeiss microscope, AxioLab A1, with an oil immersion objective 1000x. Quantitative analyses were done by calculating relative abundances by counting approximately 200 valves (up to 900 valves were counted in some levels down to 30 when diatoms were scarce) in continuous transects encompassing the whole slide. Diatoms were classified to species or variety level. Optical microscopy images (1000x) were taken using a digital SLR camera (Canon EOS Rebel) attached to microscope. The nomenclature status of species or variety was verified using the Catalogue of Diatom Names (California Academy of Sciences). Diatoms were grouped according to life forms and ecological characteristics. Data were analyzed and plotted using 'tidypaleo' version 0.1.1 packets in R 4.1.0 software.

3.3. Pollen Analysis and Macrocarbon analysis

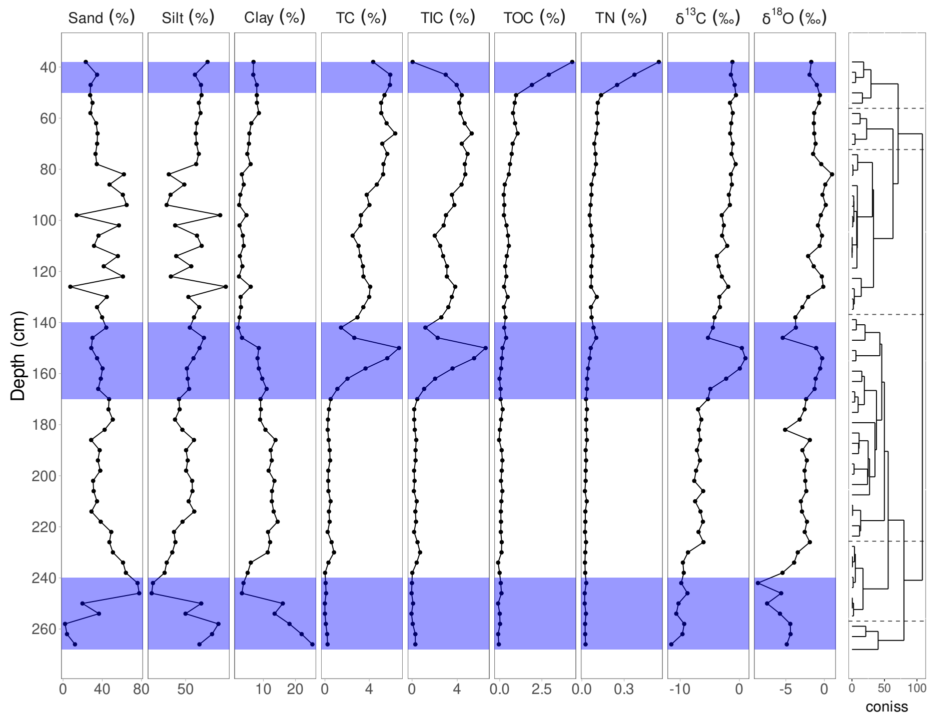
Eleven pollen samples were obtained from short core VIC11-2A-1G and processed following Faegri and Iversen (1989) methods to extract pollen grains. Lycopodium tablets were added for calculating pollen concentration (grains/cm3; Stockmarr(1971)) and accumulation rates (grains/cm2/yr). Pollen grains were mounted on glass slides and identified and quantified under an Axiostar Carl Zeiss microscope (400x and 1000x) using published pollen atlas Heusser (1971) and the reference pollen collection of the Paleoecology Laboratory at CEAZA. Pollen counts include 300 terrestrial pollen grains excluding paludal taxa. Data were analyzed and plotted using Tilia software.

3.4. Phytoliths

Phytoliths were obtained from samples of approximately 2 gr of sediment, spaced every 30 cm on the 8 long sediment cores. We followed the standard procedures for phytoliths extraction (Piperno, 1988, 1994, 2006) using 36% HCl and 70% HNO3 to remove carbonates and organic matter respectively. Phytoliths were separated by heavy liquid flotation using ZnBr2 prepared to a density of 2.3 g/cm3 (Dickau et al., 2013). Finally, phytoliths were counted until obtaining a total of 200 per sample and they were classified according to the International Code for Phytolith Nomenclature (Madella et al. 2005). The presence of maize leaf phytoliths was studied using the discriminant function of cross-shaped phytoliths (cross variants 1 and 5/6) (Piperno et al., 2006). The percentage of each family or subfamily was plotted and the zones were built through the CONISS method carried out with the Rioja package in the R software (Juggins, 2012). Additionally, a change point analysis was carried out in the sequence of arboreal phytoliths to identify possible significant increases or decreases of this vegetation cover in the studied basin (Bai and Perron, 2003; Killick and Eckley, 2014).

3.5. Chronology

The radiocarbon sample on carbon and bulk sediment were obtained of the profile exposed sampled at 1 cm resolution (Table 1) . The radiocarbon ages were determined by AMS at DirectAMS laboratories, USA. Results are presented in units of percent modern carbon (pMC) and the uncalibrated radiocarbon age before present (BP).



4. Results

4.1. Cores correlations and sedimentary proxies

References